AI ASSISTED CODING

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BATCH – 04

ASSINGMENT – 17.2

LAB – 17 : AI for Data Processing : Data Cleaning and Preprocessing

Scripts. Task – 01 : Student Academic Data Cleaning.

Prompt : Generate a Python script using Pandas to clean student academic data by handling missing values, removing duplicates, standardizing text fields, and encoding categorical variables.

Code :

TASK 01

```python
import pandas as pd
from sklearn.preprocessing import LabelEncoder

data = pd.read_csv("student_data.csv")

num_cols = data.select_dtypes(include=['int64', 'float64']).columns
data[num_cols] = data[num_cols].fillna(data[num_cols].mean())

cat_cols = data.select_dtypes(include=['object']).columns
for col in cat_cols:
    data[col].fillna(data[col].mode()[0], inplace=True)
data.drop_duplicates(inplace=True)
for col in cat_cols:
    data[col] = data[col].str.lower().str.strip()
le = LabelEncoder()
for col in cat_cols:
    data[col] = le.fit_transform(data[col])

data['marks'] = (data['G1'] + data['G2'] + data['G3']) / 3
data['attendance'] = 100 - data['absences']
print("\nCleaned Data:\n", data.head())
```

Output :

```
Cleaned Data:
    school  sex  age  address  famsize  Pstatus  Medu  Fedu  Mjob  Fjob  ...  \
0       0    0   18        1        0        0     4     4     0     4  ...
1       0    0   17        1        0        1     1     1     0     2  ...
2       0    0   15        1        1        1     1     1     0     2  ...
3       0    0   15        1        0        1     4     2     1     3  ...
4       0    0   16        1        0        1     3     3     2     2  ...

   goout  Dalc  Walc  health  absences  G1  G2  G3      marks  attendance
0      4     1     1       3         6   5   6   6   5.666667          94
1      3     1     1       3         4   5   5   6   5.333333          96
2      2     2     3       3        10   7   8  10   8.333333          90
3      2     1     1       5         2  15  14  15  14.666667          98
4      2     1     2       5         4   6  10  10   8.666667          96
```

Data Set Used : Student_data[Kaggle].

Explanation : This dataset contains multiple academic and personal features, so missing values are handled using mean and mode. Duplicate rows are removed and text fields are standardized. All categorical columns are encoded using Label encoder. New features like marks are created. The final dataset is clean and ready for analysis or ML models.

Task – 02 : Financial Transaction Data Preprocessing.

Prompt : Generate Python code to preprocess financial transaction data by converting dates, extracting time features, normalizing amounts, and handling outliers.

Code :

TASK 02

```python
import pandas as pd
from sklearn.preprocessing import MinMaxScaler

data = pd.read_csv("sample_dataset.csv")
data.rename(columns={'Date': 'transaction_date', 'Transaction Amount': 'amount'}, inplace=True)
data['transaction_date'] = pd.to_datetime(data['transaction_date'], errors='coerce')
data['month'] = data['transaction_date'].dt.month
data['year'] = data['transaction_date'].dt.year
data.ffill(inplace=True)

scaler = MinMaxScaler()
data[['amount']] = scaler.fit_transform(data[['amount']])
q1 = data['amount'].quantile(0.25)
q3 = data['amount'].quantile(0.75)
iqr = q3 - q1

lower = q1 - 1.5 * iqr
upper = q3 + 1.5 * iqr
data = data[(data['amount'] >= lower) & (data['amount'] <= upper)]

print("\nProcessed Data:\n", data.head())
```

Output :

```
Processed Data:
    Customer ID     Name    Surname Gender   Birthdate    amount  \
0       752858     Sean  Rodriguez      F  2002-10-20  0.010171
2       305449    Jacob   Williams      M  1981-10-25  0.037050
3       988259   Nathan     Snyder      M  1977-10-26  0.002104
4       764762  Crystal      Knapp      F  1951-11-02  0.019099
5       576539   Monica   Bartlett      F  2001-10-20  0.031430

  transaction_date                Merchant Name     Category  month  year
0       2023-04-03                Smith-Russell     Cosmetic      4  2023
2       2023-09-20                   Steele Inc     Clothing      9  2023
3       2023-01-11    Wilson, Wilson and Russell     Cosmetic      1  2023
4       2023-06-13                Palmer-Hinton  Electronics      6  2023
5       2023-08-24          Tran, Torres and Joyce     Cosmetic      8  2023
```

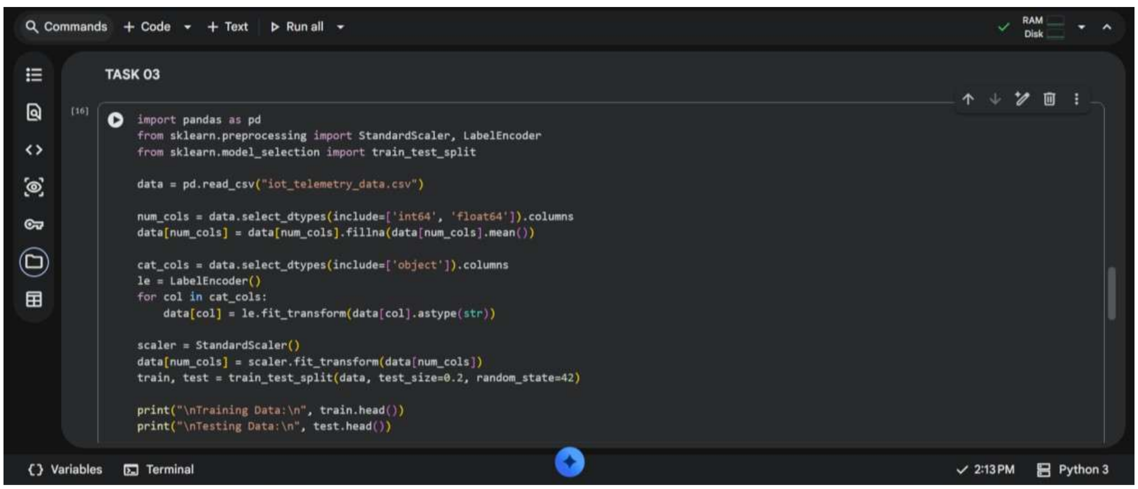
Data Set Used : Sample_data[Kaggle].

Explanation : The dataset is first loaded and the transaction date column is converted into datetime format for easier time-based analysis. New features like month and year are extracted from the date to enhance insights.
Missing values are handled using forward fill to maintain data continuity. The transaction amount is normalized using MinMaxScaler to bring values into a common range. Finally, outliers are removed using the IQR method to improve data quality and model performance.

Task – 03 : Environmental Sensor Data Preparation.

Prompt : Write Python code to clean sensor data, handle missing values, scale numerical features, encode categorical variables, and split the dataset.

Code :




```python
import pandas as pd
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.model_selection import train_test_split

data = pd.read_csv("iot_telemetry_data.csv")

num_cols = data.select_dtypes(include=['int64', 'float64']).columns
data[num_cols] = data[num_cols].fillna(data[num_cols].mean())

cat_cols = data.select_dtypes(include=['object']).columns
le = LabelEncoder()
for col in cat_cols:
    data[col] = le.fit_transform(data[col].astype(str))

scaler = StandardScaler()
data[num_cols] = scaler.fit_transform(data[num_cols])
train, test = train_test_split(data, test_size=0.2, random_state=42)

print("\nTraining Data:\n", train.head())
print("\nTesting Data:\n", test.head())
```

Data Set Used : Iot_Telemetry_Data[Kaggle].

Explanation :

The dataset is loaded and missing numerical values are filled using the mean to ensure completeness. Categorical columns are encoded into numerical form using LabelEncoder. All numerical features are standardized using StandardScaler to improve model performance. The dataset is then split into training and testing sets for machine learning tasks. This preprocessing makes the data clean, consistent, and ready for predictive modeling.

Output :

```
Training Data:
              ts  device        co  humidity  light       lpg  motion  \
403743  1.718761       2  1.049533 -0.977585  False  1.036094   False
89477  -0.965176       0  0.191110  1.239461  False  0.225863   False
285682  0.709553       1 -0.503622  0.808369   True -0.466690   False
45954  -1.339768       2  0.531880 -0.775235  False  0.552874   False
389117  1.593869       0 -1.937926  1.371428  False -2.051467   False

           smoke      temp
403743  1.039346 -0.353546
89477   0.219849 -1.242980
285682 -0.473833  1.277084
45954   0.549630 -0.131187
389117 -2.032107 -1.168861

Testing Data:
              ts  device        co  humidity  light       lpg  motion  \
362352  1.370019       1 -0.237835 -0.440919   True -0.197264   False
382697  1.540262       2  0.843570 -0.766438  False  0.845664   False
386329  1.570416       1 -0.345070  0.438861   True -0.305255   False
141680 -0.517428       0 -1.438976  1.652958  False -1.469927   False
212318  0.081979       2  0.595091 -1.162339  False  0.612720   False

           smoke      temp
362352 -0.204768  2.351818
382697  0.846025 -0.242366
```

Task – 04 : Real Time Application : Smart City Traffic Data Cleaning.

Prompt : Generate Python code to clean traffic data by standardizing text, filling missing values, removing duplicates, normalizing features, and comparing dataset quality.

Code :

TASK 04

```python
import pandas as pd
from sklearn.preprocessing import MinMaxScaler

data = pd.read_csv("smart_city_traffic_stress_dataset.csv")

num_cols = data.select_dtypes(include=['int64', 'float64']).columns
cat_cols = data.select_dtypes(include=['object']).columns

for col in cat_cols:
    data[col] = data[col].str.lower().str.strip()

for col in num_cols:
    data[col].fillna(data[col].mean(), inplace=True)

data.drop_duplicates(inplace=True)

scaler = MinMaxScaler()
data[num_cols] = scaler.fit_transform(data[num_cols])

print("\nAfter Cleaning:\n", data.describe())
```

Data Set Used : Smart_city_traffic_stress_dataset[Kaggle].

Output :

```
After Cleaning:
       traffic_density  horn_events_per_min     avg_speed  signal_wait_time  \
count     50000.000000         50000.000000  50000.000000      50000.000000
mean          0.499267             0.354817      0.530562          0.463345
std           0.291048             0.176126      0.179240          0.232271
min           0.000000             0.000000      0.000000          0.000000
25%           0.247706             0.222848      0.386131          0.264173
50%           0.495413             0.350563      0.530470          0.464173
75%           0.752294             0.479083      0.675335          0.660144
max           1.000000             1.000000      1.000000          1.000000

       road_quality_score  stress_index
count        50000.000000  50000.000000
mean             0.660046      0.444023
std              0.209512      0.163516
min              0.000000      0.000000
25%              0.516667      0.316200
50%              0.666667      0.436700
75%              0.816667      0.562700
max              1.000000      1.000000
```

Explanation :

The dataset is cleaned by standardizing text fields to ensure consistency across categorical data. Missing numerical values are filled using the mean to avoid data loss. Duplicate records are removed to improve data accuracy. Numerical features are normalized using MinMaxScaler for better comparability. The dataset summary before and after cleaning highlights improvements in data quality.

Task – 05 : Social Media Text Data Cleaning and Feature Extraction.

Prompt : Write Python code to clean social media text data by removing duplicates, cleaning text, tokenizing, removing stopwords, and extracting features.

Code :

TASK 05

```
import pandas as pd
import re
from nltk.corpus import stopwords
from nltk.stem import PorterStemmer
import nltk

data = pd.read_csv("mental_health_social_media_dataset.csv")

data.drop_duplicates(inplace=True)
text_col = data.select_dtypes(include=['object']).columns[0]
data.dropna(subset=[text_col], inplace=True)

def clean_text(text):
    text = re.sub(r"http\S+", "", str(text))
    text = re.sub(r"[^a-zA-Z]", " ", text)
    text = text.lower()
    return text
data['clean_text'] = data[text_col].apply(clean_text)
nltk.download('stopwords') # Download the stopwords corpus
stop_words = set(stopwords.words('english'))
ps = PorterStemmer()
data['tokens'] = data['clean_text'].apply(lambda x: [ps.stem(word) for word in x.split() if word not in stop_words])

data['word_count'] = data['tokens'].apply(len)
print("\nProcessed Data:\n", data.head())
```

Output :

```
Processed Data:
...        person_name  age       date  gender   platform  daily_screen_time_min  \
0     Reyansh Ghosh   35   1/1/2024    Male  Instagram                    320
1        Neha Patel   24  1/12/2024  Female  Instagram                    453
2      Ananya Naidu   26   1/6/2024    Male   Snapchat                    357
3          Neha Das   66  1/17/2024  Female   Snapchat                    190
4  Reyansh Banerjee   31  1/28/2024    Male   Snapchat                    383

   social_media_time_min  negative_interactions_count  \
0                    160                            1
1                    226                            1
2                    196                            1
3                    105                            0
4                    211                            1

   positive_interactions_count  sleep_hours  physical_activity_min  \
0                            2          7.4                     28
1                            3          6.7                     15
2                            2          7.2                     24
3                            1          8.0                     41
4                            2          7.1                     22

   anxiety_level  stress_level  mood_level mental_state        clean_text  \
0              2             7           6     Stressed     reyansh ghosh
1              3             8           5     Stressed        neha patel
2              3             7           6     Stressed      ananya naidu
3              2             6           6     Stressed          neha das
4              3             7           6     Stressed  reyansh banerjee
```

Data Set Used : Mental_health_social_media_dataset[Kaggle].

Explanation :

The dataset is cleaned by removing duplicate and missing text entries. Text data is processed by eliminating URLs, special characters, and converting everything to lowercase. Stopwords are removed and stemming is applied to simplify words. The cleaned text is tokenized into meaningful words. Additional features like word count are generated to support sentiment analysis tasks.